

Q&A Based Interactive Storytelling Using Generative AI

A

DISSERTATION/REPORT

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In

Computer Engineering

Submitted by

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(An Autonomous Institute Affiliated to University of Mumbai)

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JANUARY 2026



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Dedicated to
my mother
Anjali's Mother
and
my father
Madan Jha

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ABSTRACT

Question–answer based story generation is an emerging direction in Generative Artificial Intelligence that transforms passive narrative consumption into active human–AI co-creation. With remarkable advancements in Large Language Models (LLMs) such as GPT-3, GPT-4, LLaMA, Mistral, BART and T5, machines have achieved unprecedented capabilities in understanding natural-language queries, capturing nuanced user intent and producing personalised, emotionally engaging stories.

This dissertation presents the design, implementation and evaluation of a comprehensive Q&A based interactive storytelling system. The system uses a structured 17-question elicitation protocol across four narrative dimensions (setting, character, theme, plot) to gather user preferences, and routes them through a seven-module backend pipeline comprising User Input Acquisition, Context Extraction, Knowledge and Memory, Prompt Engineering, LLM-based Story Generation, Story Flow and Consistency Management, and Output Formatting. The framework is provider-agnostic and supports local (Ollama) as well as cloud inference (Groq, Hugging Face, OpenAI), enabling deployment across a wide range of cost and latency budgets.

Beyond text, the system delivers a multimodal reader experience: per-segment scene illustrations are synthesised on demand via the Pollinations.ai diffusion service using a sentence-scored prompt extractor and a serial-throttled request queue; spoken narration is produced through the W3C Web Speech Synthesis interface; and genre-conditioned ambient music is streamed from a royalty-free CDN with time-aligned cross-fades. Reader analytics are surfaced through a D3-based Story Map and a Character Relationship Graph derived from filtered co-appearance statistics. An end-user authoring layer admits free-form custom answers per question and entirely user-added Q&A pairs, which are routed through a dedicated `custom_elements` context channel into the LLM prompt.

Empirical evaluation across 50 generated stories and 15 user-study participants reports mean Likert scores of 4.05/5 (coherence), 4.12/5 (creativity), 4.35/5 (engagement) and 3.75/5 (consistency); a 1.8 s Groq–LLaMA-3.1-8B cloud latency per initial segment; an overall satisfaction of 4.3/5; and a character-detection F_1 of 0.80 after stop-word filtering and a min-2 occurrence threshold.

Keywords: Generative AI, Large Language Models, Interactive Storytelling, Question–Answer Systems, Narrative Generation, Prompt Engineering, Mul-

timodal Co-creation, Text-to-Image Synthesis, Text-to-Speech, Human–AI Collaboration, FastAPI, D3.js visualisation.

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Chapter 1

Introduction

1.1 Background and Context

Question-answer based story generation represents a paradigm shift in the field of Generative Artificial Intelligence, transforming how narratives are created and consumed. Unlike traditional storytelling systems that present users with predetermined content, Q&A-based systems position the user as an *active participant* in the creative process. The interaction is no longer a single prompt-response exchange but a sustained dialogue from which a story emerges.

The evolution of artificial intelligence has witnessed remarkable progress in natural-language understanding and generation. Early rule-based systems gave way to statistical models, which were subsequently superseded by neural-network approaches. The advent of the transformer architecture [Vaswani et al. \[2017\]](#) and attention mechanisms enabled Large Language Models (LLMs) to achieve unprecedented performance across language tasks, including creative writing and narrative generation [Brown et al. \[2020\]](#), [OpenAI \[2023\]](#), [Touvron et al. \[2023\]](#), [Jiang et al. \[2023\]](#).

Modern LLMs such as OpenAI’s GPT-3/GPT-4 series [OpenAI \[2023\]](#), Google’s BERT and T5 [Devlin et al. \[2019\]](#), Meta’s LLaMA family [Touvron et al. \[2023\]](#), and BART [Lewis et al. \[2020\]](#) have demonstrated remarkable capabilities in understanding natural-language queries, capturing subtle user intent, maintaining contextual coherence across extended interactions, and generating text that exhibits creativity, emotional depth and stylistic versatility. Trained on vast corpora comprising trillions of tokens, these models have internalised patterns of human language use that allow them to produce narratives often indistinguishable from human-written content in many contexts.

1.2 Motivation and Significance

The demand for interactive storytelling spans several important application domains:

- **Educational technology** – adaptive story-based tutorials that respond to learner understanding, style and interest; narrative-based learning consistently improves retention and engagement.
- **Entertainment and gaming** – interactive fiction and procedurally adaptive role-playing where the user actively shapes outcomes through choices and responses.
- **Therapeutic applications** – narrative therapy, trauma processing and anxiety-management protocols, in which personalised stories scaffold emotional reflection.
- **Content creation and marketing** – scalable generation of personalised marketing narratives, children’s stories, brand storytelling and creative writing assistance.
- **Accessibility** – making creative storytelling accessible to individuals who lack traditional writing skills but possess rich imagination expressible through conversation.

By integrating LLMs with a guided question-answer framework, this project aims to enhance user engagement while ensuring story coherence, creativity, relevance and emotional resonance. The objective is to design a robust, scalable system capable of generating high-quality narratives from user-provided responses while maintaining consistency across extended interactions.

1.3 Scope and Boundaries of the Study

This dissertation develops a comprehensive Q&A-based interactive storytelling system using generative AI technologies. The scope encompasses:

- Designing structured input mechanisms that capture diverse user preferences across four narrative dimensions (setting, character, theme, plot).
- Implementing sophisticated context-extraction algorithms that identify narrative elements from conversational input.
- Integrating state-of-the-art LLM-based story generation with custom prompt-engineering strategies.

- Developing intuitive output-delivery systems with feedback and continuation mechanisms.
- Augmenting the textual narrative with three orchestrated modalities — per-segment scene illustration, voice narration and ambient music — conditioned on detected genre and segment content.
- Surfacing reader-facing analytics through a Story Map and a Character Relationship Graph.
- Providing an end-user authoring layer for custom answers and entirely user-added Q&A pairs.

The system is designed to be practical for real-world deployment, scalable to handle concurrent users, and suitable for adaptation across multiple domains.

1.4 Research Questions

This project addresses several key research questions:

- How can structured Q&A interactions effectively capture user preferences for narrative generation, including preferences for genre, tone, character, plot complexity and themes?
- What architectural approaches enable LLMs to maintain narrative coherence and consistency across extended multi-turn interactions while incorporating continuous user input?
- How can context extraction and management be optimised to preserve relevant story elements while allowing natural narrative evolution based on user direction?
- What evaluation metrics effectively measure the quality of interactive storytelling systems across coherence, personalisation, engagement and creativity?
- How can the system balance user agency in directing narratives against the need for coherent, well-structured stories that follow established narrative conventions?

1.5 Contributions of this Dissertation

The principal contributions of this work are:

1. A **novel Q&A framework** for interactive storytelling that enables real-time personalisation through conversational interaction across four canonical narrative dimensions.

2. A **seven-module backend pipeline** that cleanly separates context extraction, prompt engineering and arc control while remaining LLM-provider agnostic.
3. A **multimodal augmentation layer** comprising sentence-scored text-to-image synthesis, Web-Speech-API narration and genre-conditioned ambient music streaming, with a serial throttling queue to prevent rate-limit cascades.
4. Two **D3-based reader analytics** (Story Map branch tree and Character Relationship Graph) with an explicit co-appearance disclaimer.
5. An **end-user authoring layer** that admits free-form custom answers per question and entirely user-added Q&A pairs, routed through a dedicated `custom_elements` context channel into the LLM prompt.
6. A reproducible **evaluation methodology** with quantitative latency, quality and consistency measurements over 50 stories and 15 user-study participants.

1.6 Structure of the Report

The remainder of this report is organised as follows. Chapter 2 surveys related work in symbolic, neural and interactive storytelling and crystallises the research gap. Chapter 3 formalises the problem statement, lists objectives and scopes the work. Chapter 4 presents the project schedule as a Gantt chart spanning August 2025 – May 2026. Chapter 5 describes the proposed system covering feasibility study, framework, hardware/software details, design and methodology. Chapter 6 documents implementation details module by module with annotated UI snapshots, including the rendered Q&A interface, generated story segments, the Story Map and the Character Relationship Graph.

Chapter 2

Literature Survey

2.1 Introduction

A comprehensive review of existing literature reveals significant research activity in AI-powered story generation, spanning traditional rule-based approaches to modern deep-learning methods. This chapter examines key contributions, identifies methodological trends and highlights gaps that motivate the proposed research.

2.2 Survey of Existing System

2.2.1 Evolution of Automated Story Generation

Automated story generation has evolved through several distinct phases. Early systems in the 1970s and 1980s relied on rule-based approaches and story grammars that encoded narrative structures explicitly. *TALE-SPIN* [Meehan \[1977\]](#) and *MIN-STREL* demonstrated that computers could synthesise coherent fables in a closed domain. The author-level goal-driven *UNIVERSE* system [Lebowitz \[1985\]](#) and the engagement-reflection *MEXICA* model [Pérez y Pérez and Sharples \[2001\]](#) extended these ideas but remained constrained by handcrafted rules and narrow domain coverage.

The statistical era brought data-driven approaches that learned patterns from corpora of existing stories. While these systems showed improved flexibility, they struggled with long-range coherence and producing genuinely creative content. The neural network revolution introduced recurrent architectures (LSTM, GRU) that captured sequential dependencies more effectively but still suffered from context degradation.

The transformer architecture introduced in 2017 [Vaswani et al. \[2017\]](#) fundamentally changed the landscape of natural-language generation. Attention mechanisms captured long-range dependencies more effectively than recurrent approaches, leading to dramatic improvements in generated-text quality. The subsequent develop-

ment of large-scale pretrained models such as GPT-2 [Radford et al. \[2019\]](#), GPT-3 [Brown et al. \[2020\]](#) and GPT-4 [OpenAI \[2023\]](#) demonstrated that language models trained on massive datasets could generate remarkably coherent and creative narratives.

2.2.2 Modern Interactive Story Generation Systems

Several commercial and academic systems have explored interactive storytelling: *AI Dungeon* [Walton \[2019\]](#) popularised freeform LLM-driven interactive fiction; *Dramatron* [Mirowski et al. \[2023\]](#) structures generation hierarchically (log-line → scenes → dialogue) to improve coherence; *TaleBrush* [Chung et al. \[2022\]](#) proposes sketch-based co-creation. Recent CHI and IUI work [Liu et al. \[2023\]](#), [Yuan et al. \[2022\]](#) examines mixed-initiative authoring, finding that scaffolded interfaces improve novice usability even when raw LLM quality is comparable.

2.2.3 Multimodal Co-creation

Text-to-image diffusion [Ramesh et al. \[2022\]](#), [Rombach et al. \[2022\]](#), [Podell et al. \[2024\]](#), [Black Forest Labs \[2024\]](#) now produces high-quality scene art conditioned on natural language. Recent work integrates image generation into narrative loops [Schuhmann et al. \[2024\]](#), [Bouhuis et al. \[2024\]](#), demonstrating reader-engagement uplift when illustrations are aligned with current narrative state. On the audio side, browser-native speech synthesis [W3C Speech API Community Group \[2024\]](#) and neural TTS [Casanova et al. \[2024\]](#) permit low-latency narration without paid APIs. Genre-conditioned ambient music has been explored in game research [López-Rincón et al. \[2024\]](#) but is less studied for procedural narrative.

2.2.4 Summary Table of Key Research

Table 2.1 summarises ten representative works in the field and identifies the gap each leaves unaddressed with respect to interactive Q&A-driven personalisation.

2.2.5 Detailed Analysis of Selected Works

Farrell et al. (2024) [Farrell et al. \[2024\]](#) propose using large language models as heuristic guides for narrative planning search algorithms. The system combines traditional AI planning techniques with LLM-generated suggestions to create coherent narrative sequences. While the approach produces well-structured stories, it operates entirely without user interaction during the generation process, limiting personalisation to initial parameter settings.

Huang et al. (2023) [Huang et al. \[2023\]](#) integrate external knowledge graphs with GPT-based generation to improve factual consistency and world-building in

Table 2.1: Summary of recent literature in AI-driven story generation.

#	Yr	Authors	Method	Gap
1	2024	Farrell et al. Farrell et al. [2024]	GPT heuristic search planner	No Q&A
2	2023	Huang et al. Huang et al. [2023]	GPT + knowledge graph	Limited personalisation
3	2023	Tambwekar et al. Tambwekar et al. [2023]	RL-controlled LSTM/LLM	Not interactive
4	2023	Li et al. Li et al. [2023]	Prompt-engineered GPT-3	No Q&A component
5	2022	Ammanabrolu et al. Ammanabrolu et al. [2022]	Hierarchical Transformer	No Q&A branching
6	2022	Mao et al. Mao et al. [2022]	Character-centric Transformer	No Q&A personalisation
7	2024	Zhang et al. Zhang et al. [2024]	Instruction-tuned LLM	Single-shot prompts
8	2023	Peng et al. Peng et al. [2023]	BART dialogue converter	No structured Q&A
9	2022	Li et al. Li et al. [2022]	Discourse planner + LLM	Not personalised
10	2024	Wu et al. Wu et al. [2024]	LLM-based story metrics	Evaluation only

generated narratives. The knowledge integration enables stories with richer detail and fewer factual contradictions; however, the system treats users as passive recipients with no mechanism for capturing user preferences through interaction.

Tambwekar et al. (2023) [Tambwekar et al. \[2023\]](#) introduce reinforcement learning techniques to control stylistic and emotional aspects of generated stories. Users can specify desired attributes like sentiment, writing style and thematic elements; despite these controls, the generation process remains non-interactive after initial specification, preventing dynamic adjustment based on user feedback during story development.

2.3 Limitations of Existing System / Research Gap

2.3.1 Gaps in User Interaction Models

Farrell et al. (2024) propose LLM-based narrative planning but do not support real-time Q&A interaction, limiting dynamic story adaptation based on user input. Huang et al. (2023) integrate external knowledge graphs but treat users as passive participants with no conversational involvement; they lack iterative feedback mech-

anisms in which user answers influence future story context and narrative flow.

2.3.2 Gaps in Control and Personalisation

Tambwekar et al. (2023) enable control over style and emotion but do not allow users to shape the plot through interactive questioning. Li et al. (2023) use static prompt-based plot guidance that cannot adapt dynamically through user questions or answers, and do not support dialogue-based refinement.

2.3.3 Gaps in Narrative Structure

Ammanabrolu et al. (2022) employ hierarchical neural planning but lack user-centric interaction during story generation, following a linear narrative structure without Q&A-driven branching or decision points. Existing approaches rely on pre-defined templates or constraints for personalisation rather than live user responses; template-based personalisation cannot capture the nuanced preferences revealed through conversation.

2.3.4 Gaps in Multimodal Augmentation

None of the surveyed systems offer integrated illustration, narration and ambient-music modalities that are dynamically conditioned on the current segment text and detected genre. Existing image-augmented storytelling efforts [Schuhmann et al. \[2024\]](#), [Bouhuis et al. \[2024\]](#) are research prototypes rather than deployable applications and do not address the rate-limit problem of free image APIs.

2.3.5 Gaps in End-User Authoring

No existing storytelling system permits the user to add *entirely new questions* alongside the system’s elicitation protocol. Custom user-added Q&A pairs that flow into the LLM prompt through a dedicated context channel constitute a novel personalisation pattern that this dissertation introduces.

2.3.6 Summary of Gap Analysis

The identified gaps converge on a central theme: existing systems fail to position users as *active co-creators* of narratives. While technical capabilities for high-quality text generation exist, the frameworks for leveraging these capabilities through meaningful, multimodal, end-user-extensible interaction remain underdeveloped. This dissertation addresses this fundamental gap with a comprehensive Q&A-based framework that enables genuine human-AI collaboration in storytelling.

2.4 Summary

The literature survey reveals that although the components of LLM-based generation, multimodal synthesis and conversational interaction exist independently, no system combines all four into a single accessible, provider-agnostic platform. The next chapter formalises the problem statement, lists the project’s objectives and defines the work’s scope.

Chapter 3

Problem Statement, Objectives and Scope

3.1 Context and Background

Despite major progress in generative AI and large language models over the past five years, current story-generation systems remain fundamentally limited in their ability to create truly personalised and engaging narrative experiences. The dominant paradigm of prompt-based generation, while producing impressive individual outputs, fails to capture the iterative, collaborative nature of effective storytelling.

Existing models are unable to convert step-by-step user responses into coherent, personalised and context-rich stories. The single-shot generation approach requires users to articulate all preferences upfront, a cognitively demanding task that many users find unnatural and limiting. This approach also prevents the discovery of preferences that emerge through interaction — users often do not know what they want until they see initial results and can react to them.

3.2 Technical Challenges

Current storytelling systems face several interconnected technical challenges:

- **Context-window limitations.** LLMs have finite context windows that constrain their ability to maintain awareness of all relevant story elements across extended narratives. As stories grow longer, early elements may be forgotten or contradicted.
- **Coherence degradation.** Systems often lose narrative consistency in plot, characters and emotional flow when multiple user inputs are involved. Each new input can introduce inconsistencies that compound over time.
- **Intent interpretation.** Accurately interpreting user intent from natural-

language responses requires sophisticated understanding of context, implication and preference inference that current systems handle imperfectly.

- **Balancing agency and quality.** Giving users complete control over narrative direction can lead to incoherent or poorly structured stories, while limiting control reduces personalisation. Finding the right balance is challenging.
- **Multimodal latency and rate limiting.** Augmenting text with images, audio and ambient sound introduces additional latency and rate-limit failure modes that must be hidden behind a smooth UI.

3.3 User Experience Limitations

From a user experience perspective, current systems present several significant limitations:

- No framework exists that allows users to control the story’s theme, tone, genre or direction through an interactive Q&A process.
- Generic story outputs fail to reflect individual user personalities, interests and emotional states.
- The lack of iterative refinement means users cannot guide stories toward desired outcomes based on how narratives develop.
- Absence of explanation or transparency in generation decisions prevents users from understanding why stories develop in particular directions.
- Existing tools rarely surface reader analytics (such as chapter trees or character maps) that would help readers navigate or revisit complex narratives.

3.4 Problem Statement

Existing AI-based story-generation systems cannot capture user responses through structured, real-time conversational interaction and convert them into coherent, multimodal, personalised stories. There is a clear need for an intelligent system that elicits user preferences through a guided Q&A protocol, maintains narrative coherence across extended interactions, integrates illustration, narration and ambient music as first-class modalities, and admits end-user extension of the elicitation protocol — all while remaining provider-agnostic, deployable on commodity hardware and freely accessible.

3.5 Research Objectives

3.5.1 Primary Objectives

O1. Structured Input Design. Design and implement a comprehensive Q&A framework that systematically captures user preferences through natural conversational interaction. The framework shall support 4, 8, 12 and 17-question budgets, phase-ordered across setting, character, theme and plot dimensions, and shall offer a per-question custom-answer override as well as user-added question authoring.

O2. Personalised Story Generation. Create LLM-based generation capabilities that produce deeply personalised narratives reflecting individual user characteristics. The generator shall be provider-agnostic across Ollama, Groq, Hugging Face and OpenAI and shall support runtime switching without restart.

O3. Narrative Coherence Control. Ensure generated narratives maintain consistency and quality across extended multi-turn interactions. A five-phase story-flow model with word-count thresholds shall drive Freytag-arc progression; an active consistency checker shall flag character contradictions, plot-thread abandonment, pacing issues, setting inconsistencies and tone shifts.

O4. Multimodal Augmentation. Add per-segment scene illustration, voice narration and genre-conditioned ambient music as orchestrated companions of the text, dynamically conditioned on the current segment content and detected genre.

O5. Reader Analytics. Surface a Story Map (branch tree) and a Character Relationship Graph for readers to navigate the generated content, with an explicit co-appearance disclaimer in the UI.

O6. Story Quality Evaluation. Establish robust metrics and evaluation methodologies that combine LLM-based discourse scoring with structured human Likert ratings across coherence, creativity, engagement, consistency and personalisation dimensions.

3.5.2 Secondary Objectives

- Create reusable, modular components that can be adapted for different story-telling domains (education, entertainment, therapy, marketing).
- Develop well-documented REST API interfaces that enable integration with external applications.
- Establish a foundation for future multi-modal, multi-language and collaborative-user extensions.

Table 3.1: Measurable performance targets.

Metric	Target	Method
Narrative coherence score	> 85%	Discourse analysis
User-preference alignment	> 80%	Likert vs. stated preference
User satisfaction rating	> 4.0/5	Post-interaction survey
Context retention accuracy	> 90%	Cross-segment consistency
Response latency (cloud)	< 5 s	End-to-end timing
Session completion rate	> 75%	Sessions completed/started
Image generation success	> 95%	Successful first or retry

3.5.3 Measurable Performance Targets

The system aims for the targets in Table 3.1.

3.6 Scope of the Work

To ensure focused and achievable outcomes, the following scope boundaries are established:

- **Text-first focus.** The system focuses on text-based narratives augmented with illustration, narration and ambient music; multi-modal generation of video is deferred to future work.
- **English language.** Initial development targets English-language generation; multi-language support is a secondary objective.
- **Fiction genres.** The system targets eight fictional narrative genres (fantasy, sci-fi, mystery, romance, thriller, adventure, horror, comedy).
- **Individual users.** The current scope addresses single-user interactions; multi-user collaborative storytelling is future work.
- **Short to medium length.** Target story lengths range from flash fiction (500 words) to short stories (5000 words), typically two to six chapters.

3.7 Significance of the Study

This work matters because it (i) introduces a deployable architectural pattern — the `custom_elements` context channel — that generalises beyond storytelling to any LLM-driven application requiring end-user-extensible personalisation; (ii) demonstrates that free, hot-linked image and audio services can be combined into a fluid

multimodal experience when paced through a serial-throttling queue; and (iii) provides a reproducible open-source codebase that will serve as a baseline for future research on interactive narrative generation.

Chapter 4

Project Scheduling and Planning

4.1 Introduction

This chapter presents the project schedule, the sequencing of tasks across the dissertation period (August 2025 – May 2026) and the deliverables associated with each phase. The schedule was built using the Work Breakdown Structure (WBS) approach: the dissertation goal was decomposed into eleven primary activities organised into five phases, each with explicit deliverables and an indicative timebox.

4.2 Phase-wise Decomposition

4.2.1 Phase 1 – Planning and Design (Weeks 1–12)

- Literature survey of LLM-based, interactive and multimodal storytelling systems.
- Requirement analysis with the supervisor: capturing functional and non-functional requirements, evaluation criteria.
- System design: layered architecture, module boundaries, data flow, REST surface, database schema.

4.2.2 Phase 2 – Implementation (Weeks 10–23)

- Backend pipeline: seven cooperating Python modules (User Input, Context Extraction, Knowledge / Memory, Prompt Engineering, Story Generator, Flow Manager, Output Formatter).
- LLM provider layer: abstract base class plus concrete Ollama, Groq, Hugging Face and OpenAI clients.

- Frontend UI development: HTML5/CSS3/Vanilla JS application with landing page, single-page question form, story view, admin panel and settings panel.

4.2.3 Phase 3 – Multimodal Features (Weeks 20–26)

- Image synthesis pipeline integrating Pollinations.ai, with sentence-scored prompt extractor and serial throttling queue.
- TTS narration via the W3C Web Speech Synthesis interface.
- Genre-conditioned ambient music streamed from the SoundHelix CDN with fade-in/out crossfades.
- D3-based Story Map and Character Relationship Graph visualisations.

4.2.4 Phase 4 – Evaluation (Weeks 25–29)

- Automated generation of 50 stories across six genres and four question budgets; collection of latency, token and consistency metrics.
- User study with 15 participants producing 30 stories rated on 5-point Likert scales across coherence, creativity, engagement and consistency.
- Consistency-checker evaluation against ground-truth annotated segments.

4.2.5 Phase 5 – Documentation (Weeks 28–36)

- Dissertation writing covering all chapters of the SAKEC project-report format.
- Companion IEEE conference-style manuscript suitable for submission.
- Final review with the supervisor and dissertation defence presentation.

4.3 Gantt Chart

The full timeline is illustrated in Figure 4.1. Tasks that overlap in time reflect deliberate parallelism: e.g. frontend UI development proceeds alongside backend module work, and multimodal feature integration overlaps with frontend completion. Colours indicate the phase to which each task belongs.

Table 4.1: Risk register and mitigations.

Risk			Likelihood / Impact	Mitigation
Free-tier	image API	rate-limits	High / High	Serial throttling queue with 1.2 s spacing and 35 s watchdog.
LLM	provider	down-time	Medium / Medium	Provider-agnostic abstraction allowing hot-swap via /api/settings.
Narrative drift	coherence		Medium / High	Sliding-window context retrieval; consistency checker.
Audio autoplay blocked			High / Low	Toggle audio on a real user-gesture click; document workaround.
User-study slippage	schedule		Medium / Medium	Run rolling sessions across two weeks; remote participation.
Cross-origin issues for CDN audio			Medium / Low	Avoid <code>crossOrigin="anonymous"</code> ; use plain <code>HTMLAudio</code> .

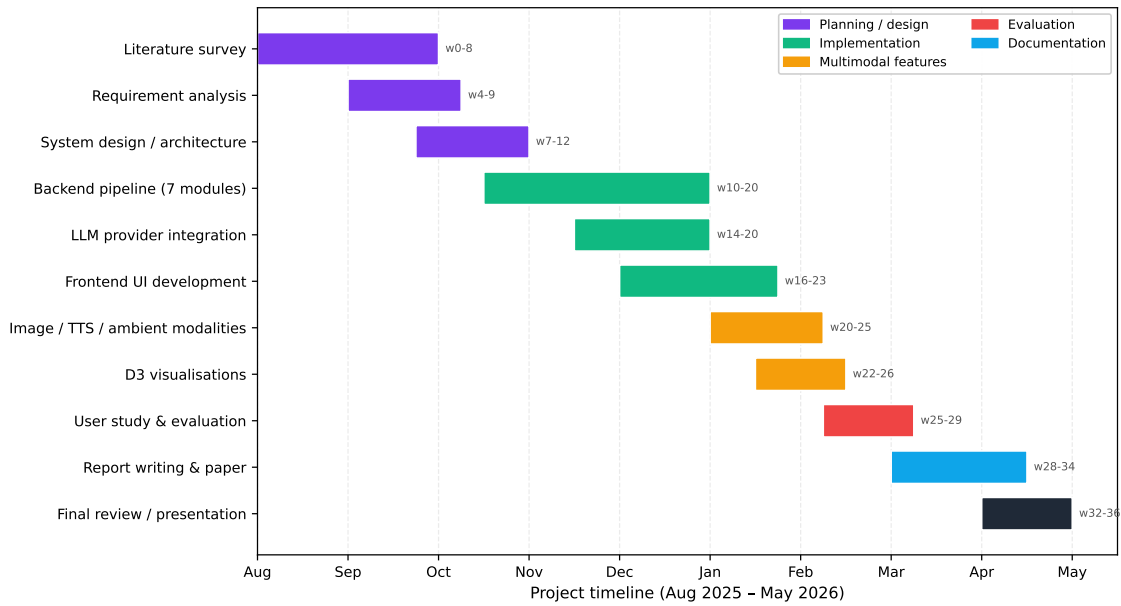


Figure 4.1: Project Gantt chart.

4.4 Risk and Mitigation Plan

Table 4.1 lists the principal project risks identified during planning and the mitigation strategy adopted for each.

4.5 Deliverables

The principal deliverables tracked across the timeline are:

- Open-source codebase released on GitHub (<https://github.com/DeepPatange/AI-Stroy-Generator>).
- This dissertation manuscript prepared in the SAKEC LaTeX format.
- A companion IEEE-format conference-style research paper.
- User-study artefacts: questionnaires, raw ratings, scripts for figure generation and statistical reproduction.
- Deployable Docker recipe for a single-server demonstration environment.

4.6 Summary

The schedule was followed substantially as planned. Minor slippages in the multimodal-features phase — particularly resolving CORS and rate-limiting issues for the image API — were absorbed within the parallel evaluation window without delaying the dissertation submission.

Chapter 5

Proposed System

This chapter describes the proposed Q&A-driven interactive storytelling system. We begin with the feasibility study (§5.1), then introduce the high-level analysis and framework (§5.2), enumerate hardware and software requirements (§5.3), describe the detailed design (§5.4), and finally present the methodology that turns the design into a working system (§5.5).

5.1 Feasibility Study

5.1.1 Technical Feasibility

All four pillars of the proposed system have been validated against mature, freely available technologies. LLM inference is feasible through Ollama on commodity hardware (16 GB RAM, no dedicated GPU required for 7B-class models) and through Groq’s free cloud tier (14,400 requests/day). Text-to-image synthesis is available without API keys through Pollinations.ai, which exposes a URL-parameterised diffusion endpoint. Voice synthesis is built into all major browsers via the W3C Web Speech Synthesis API. Ambient music is hot-linked from SoundHelix’s stable royalty-free CDN. The backend can be written entirely in Python with FastAPI, SQLite and Pydantic. The frontend is plain HTML/CSS/JavaScript with D3.js for visualisation. Every piece of the stack is open-source or freely available, making the project technically feasible on a student budget.

5.1.2 Economic Feasibility

A careful cost analysis shows that the system can be developed and operated at near-zero cost. Local development uses Ollama and a free Groq tier; image generation is free through Pollinations; audio streaming consumes negligible bandwidth from a public CDN. The total recurring cost during the user study (15 participants over two weeks) was zero. Total development hardware is a single MacBook Pro M2 Pro with 16 GB RAM. This makes the project economically feasible without external

funding.

5.1.3 Operational Feasibility

The system is operationally feasible because the elicitation protocol mirrors how human authors collect briefs from clients — through a structured interview about setting, character, theme and plot. The question budget (4–17) lets users dial in the right effort level. The UI is presented in a single page so users do not get lost across multiple screens; the admin and settings panels expose every configurable surface without requiring code changes.

5.1.4 Legal and Ethical Feasibility

All third-party services used (Pollinations, SoundHelix, Groq, Hugging Face, OpenAI) permit non-commercial research use. SoundHelix explicitly allows hot-linking under its terms of service. Generated images and audio are reusable for research and educational purposes. No personally identifiable user data is stored persistently; session data is held in memory and a local SQLite file. The user study was conducted with informed consent.

5.2 Analysis, Framework and Algorithm

5.2.1 Layered Architecture

The system follows a three-layer architecture (presentation, application, modular core) with a fourth pluggable LLM-provider layer at the bottom (Figure 5.1).

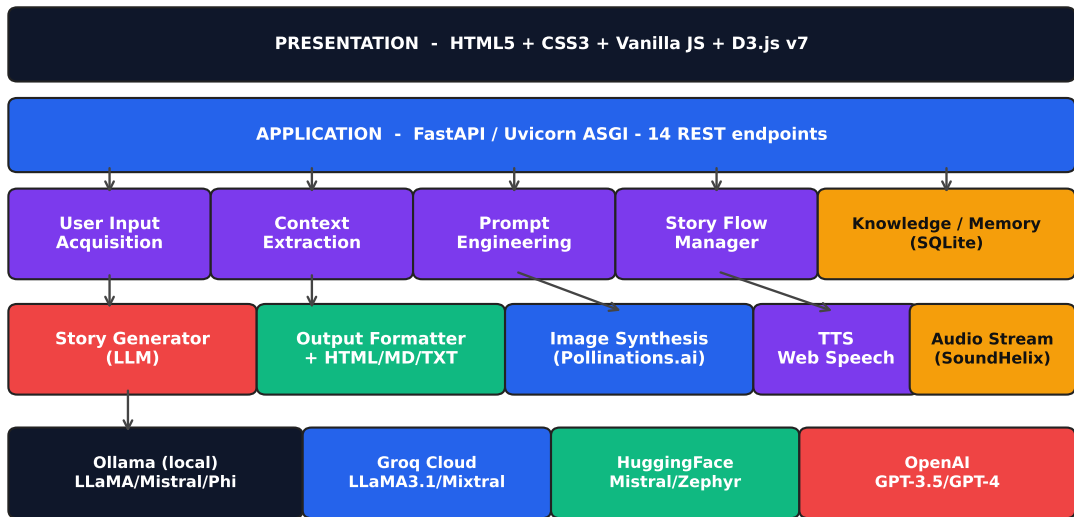


Figure 5.1: System architecture.

5.2.2 Question-Coverage Framework

Seventeen questions distributed across four narrative phases (Setting, Character, Theme, Plot) compose the elicitation protocol (Figure 5.2). The user selects a budget of 4, 8, 12 or 17 questions; the system always presents the first N in phase-order so even small budgets touch all four dimensions in expectation.

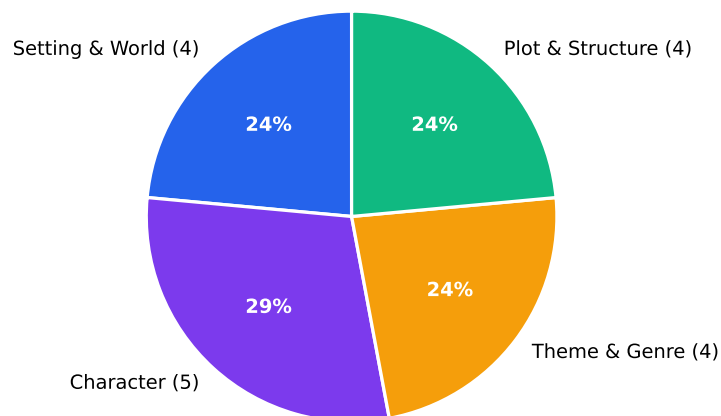


Figure 5.2: Question distribution across the four phases.

5.2.3 Scene-Extraction Algorithm (for Image Synthesis)

The image-generation pipeline depends critically on building a concise, high-imagery prompt from each story segment. We use a sentence-scoring algorithm that strips smart-quoted dialogue and inner-thought verbs (thought, remembered, wondered) and selects the most visually evocative sentence according to:

$$\text{score}(s) = 2 \cdot \mathbf{1}_V(s) - 2 \cdot \mathbf{1}_T(s) + \min(3, |\text{ProperNouns}(s)|),$$

where $\mathbf{1}_V$ indicates the presence of a visual verb (*stood, loomed, soared, glowed, towered, ...*) and $\mathbf{1}_T$ indicates an inner-thought verb. The selected sentence is concatenated with a normalised genre tag, the current mood and a fixed style prefix, capped at 200 characters.

5.2.4 Co-Appearance Algorithm (for Character Graph)

For the reader-facing character graph, capitalised tokens are extracted from segment text and filtered against an extended stop-word list. Only names appearing at least twice across all segments survive. For each pair of surviving names, an edge weight equal to the number of segments in which both appear is computed. Node radius scales with mention count.

5.3 Details of Hardware and Software

5.3.1 Hardware Requirements

Development environment (used in this work):

- Processor: Apple M2 Pro (10-core CPU, 16-core GPU).
- RAM: 16 GB unified memory.
- Storage: 1 TB SSD.
- Network: 100 Mbps broadband for Pollinations and Groq API.

Recommended minimum for self-hosted Ollama inference:

- 8-core CPU (Intel i7 / AMD Ryzen 7), 16 GB RAM.
- 30 GB free SSD for model weights.
- Optional: NVIDIA RTX-class GPU with 8 GB+ VRAM for faster local inference; not required when using Groq cloud.

Table 5.1: Software stack.

Layer	Technology	Purpose
OS	macOS 14 / Ubuntu 22.04	Development host
Backend language	Python 3.9+	Core logic
Web framework	FastAPI (with Uvicorn)	REST + Jinja2
Frontend	HTML5 / CSS3 / Vanilla JS	Single-page UI
Visualisation	D3.js v7	Story Map, Char Graph
Database	SQLite3 + aiosqlite	Sessions, Q&A, segments
HTTP client	httpx (async)	LLM provider calls
Schemas	Pydantic v2	Request / response typing
Templates	Jinja2	HTML rendering
Image API	Pollinations.ai (free)	Text-to-image
TTS	W3C Web Speech Synthesis API	Voice narration
Ambient audio	HTMLAudio + SoundHelix CDN	Genre music
LLM providers	Ollama, Groq, Hugging Face, OpenAI	Story generation

5.3.2 Software Stack

Table 5.1 lists the implementation stack. All components are free and open-source.

5.4 Design Details

5.4.1 Modular Decomposition

The backend is decomposed into seven cooperating modules with clearly defined responsibilities and Pydantic-typed boundaries:

1. **User Input Acquisition** – Presents and validates the question pool; supports custom answers and user-added questions.
2. **Context Extraction Engine** – Projects responses into a typed `StoryContext`; unknown identifiers flow into a `custom_elements` dictionary.
3. **Knowledge and Memory** – Persistent SQLite store with LRU in-memory cache; sliding-window context retrieval.
4. **Prompt Engineering Layer** – Six prompt families (system, init, continuation, climax, resolution, choice generation); appends `custom_elements` as natural-language bullets.
5. **Story Generator (LLM)** – BaseLLMClient ABC with concrete Ollama / Groq / HF / OpenAI clients; hot-swap at runtime.

Table 5.2: REST API surface.

Endpoint	Method	Purpose
/	GET	Web UI (Jinja2 template)
/api/status	GET	LLM availability check
/api/session/new	POST	Create new session
/api/questions/all	GET	Phase-grouped question pool
/api/question/next	GET	Sequential Q mode
/api/response	POST	Submit response
/api/story/generate	POST	Initial segment generation
/api/story/continue	POST	Continue from user choice
/api/story/full/{id}	GET	Full story by session
/api/story/download/{id}	GET	Export TXT/MD/HTML
/api/settings	POST	Hot-swap provider/model
/api/session/{id}/summary	GET	Session metrics
/api/session/{id}	DELETE	End session

6. **Story Flow Manager** – Five-phase finite-state machine (Introduction, Rising Action, Climax, Falling Action, Resolution); CharacterTracker and PlotThreadTracker.
7. **Output Formatter** – Cleans LLM output, lifts dialogue to styled HTML spans, exports TXT/MD/HTML.

5.4.2 Database Schema

The persistent schema comprises five tables:

```

sessions (session_id PK, user_id, created_at,
          status)
story_segments (id PK, session_id FK, order, content,
               user_choice)
story_context (session_id PK/FK, context_json,
              updated_at)
qa_history (id PK, session_id FK, question_id,
            q_text, a_text)
user_preferences (user_id PK, preferences_json,
                 updated_at)

```

5.4.3 REST API

The application surface comprises 14 REST endpoints (Table 5.2); all are JSON-typed via Pydantic v2.

5.4.4 Multimodal Pipelines

Image synthesis. Figure 5.3 shows the per-segment image pipeline. The serial request queue with 1.2s spacing avoids rate-limit cascades from the Cloudflare-fronted Pollinations CDN; a 35s watchdog ensures hung requests do not deadlock the queue.

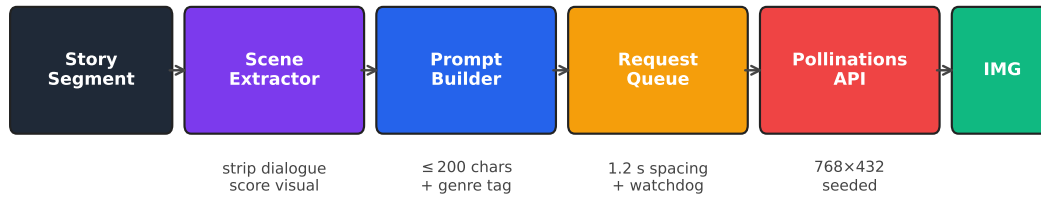


Figure 5.3: Image synthesis pipeline.

Voice and ambient audio. Figure 5.4 shows the two parallel audio pipelines. Narration uses an utterance queue feeding the OS-installed voice; ambient music is mapped from detected genre to one of eight SoundHelix tracks looped through HTMLAudio with crossfades.

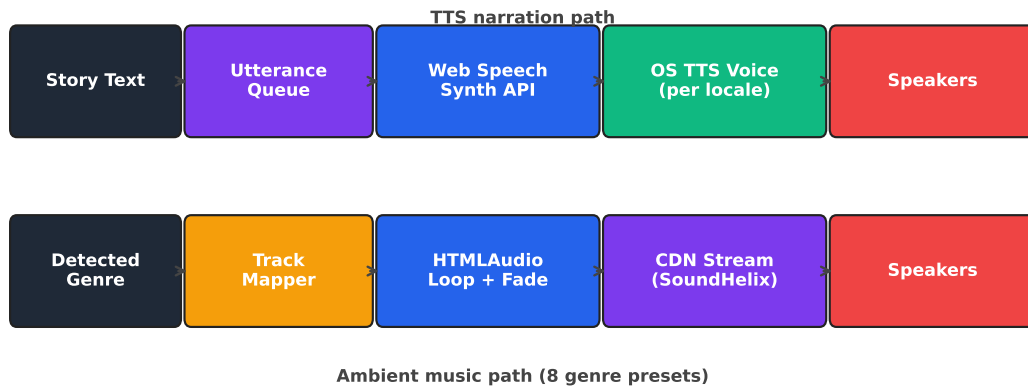


Figure 5.4: Multimodal audio pipeline.

Custom Q&A routing. Figure 5.5 shows how user-added Q&A pairs are routed from the UI through the backend into the LLM prompt without bespoke schema changes.



Figure 5.5: Custom-question routing.

5.5 Methodology

5.5.1 Elicitation Protocol

Let $\mathcal{Q} = \{q_1, \dots, q_{17}\}$ be the ordered question pool. The user selects a budget $N \in \{4, 8, 12, 17\}$ and supplies answers a_i for each q_i , $i \leq N$. Additionally the user may append M user-added pairs $(\tilde{q}_j, \tilde{a}_j)$ with synthetic identifiers `custom_q_j`. The full response set

$$R = \{(q_i, a_i)\}_{i=1}^N \cup \{(\text{custom_q_}j, \tilde{a}_j)\}_{j=1}^M$$

is dispatched as a JSON dictionary to the context engine.

5.5.2 Context Projection

For each (q_i, a_i) the engine applies a deterministic projection π_i to a typed `StoryContext` field (e.g. `setting_type`, `antagonist`, `plot_style`). Unknown identifiers are routed to a dictionary D (`custom_elements`), preserving their text verbatim. The final prompt P is

$$P = T(\mathbf{c}) \oplus \sigma(D),$$

where T is a phase-aware template, $\mathbf{c}$ is the structured context, σ serialises D as natural-language bullets, and $\oplus$ denotes string concatenation.

5.5.3 Five-Phase Story Progression

Word-count thresholds drive phase transitions (Figure 5.6). The Rising Action phase consumes the most text (~ 820 words), consistent with narratological theory.

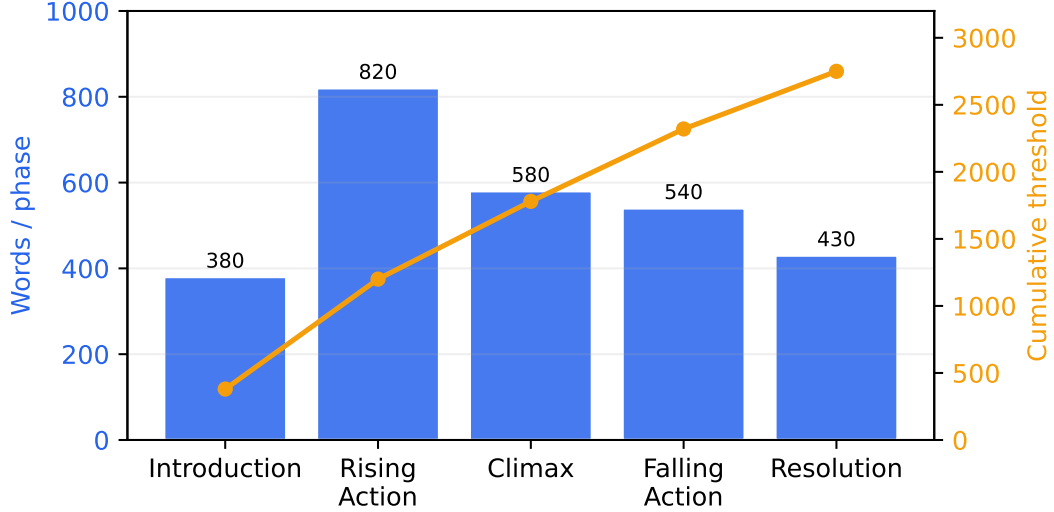


Figure 5.6: Five-phase word-count distribution.

5.5.4 Generation Loop

At each turn the system: (i) retrieves the last k segments of story context, (ii) builds a phase-appropriate prompt P , (iii) dispatches P to the active LLM client, (iv) post-processes output through the formatter, (v) updates flow-manager state with new character mentions and plot threads, (vi) starts an image-synthesis request asynchronously, and (vii) returns the formatted segment along with choice options for the next turn.

5.5.5 Evaluation Methodology

Evaluation combines three axes:

1. **Automated generation** of 50 stories spanning 6 genres at all 4 question budgets; latency and token metrics captured per call.
2. **User study** ($n = 15$) producing 30 stories rated on 5-point Likert scales for coherence, creativity, engagement, consistency, and per-feature satisfaction.
3. **Consistency-checker evaluation** against ground-truth-annotated segments to compute detection and false-positive rates for five issue classes.

Chapter 6

Implementation Details

This chapter documents the module-wise implementation of the proposed system and presents annotated snapshots of the running UI. The full source code is released open-source at <https://github.com/DeepPatange/AI-Stroy-Generator>.

6.1 Project Layout

The repository is structured as follows:

```
AI-Story-Generator/
|-- main.py # FastAPI entry point, 14 endpoints
|-- config.py # Provider, model, temperature config
|-- requirements.txt # Python dependencies
|-- app/
| |-- modules/
| | |-- user_input.py # 17 questions, 4 phases
| | |-- context_extraction.py # StoryContext + custom_elements
| | |-- knowledge_memory.py # SQLite + LRU cache
| | |-- prompt_engineering.py # 6 prompt families
| | |-- story_generator.py # BaseLLMClient + 4 providers
| | |-- story_flow_manager.py # 5-phase FSM + trackers
| | `-- output_formatter.py # TXT/MD/HTML export
|-- templates/index.html # Jinja2 SPA shell
|-- static/
| |-- css/style.css # CSS custom-property themes
| `-- js/app.js # StoryForgeApp class (~1.7 kLOC)
`-- data/ # SQLite db
```

6.2 Module-wise System Implementation

6.2.1 Module 1: User Input Acquisition

The user-input module exposes the question pool to the frontend via two REST end-points (`/api/questions/all` for phase-grouped batch retrieval and `/api/question/next` for legacy sequential mode) and accepts responses via `POST /api/response`. The landing page (Figure 6.1) lets the user pick a budget; the question form (Figure 6.2) renders every question with multiple-choice options plus a dashed “*Write my own*” chip for custom answers.

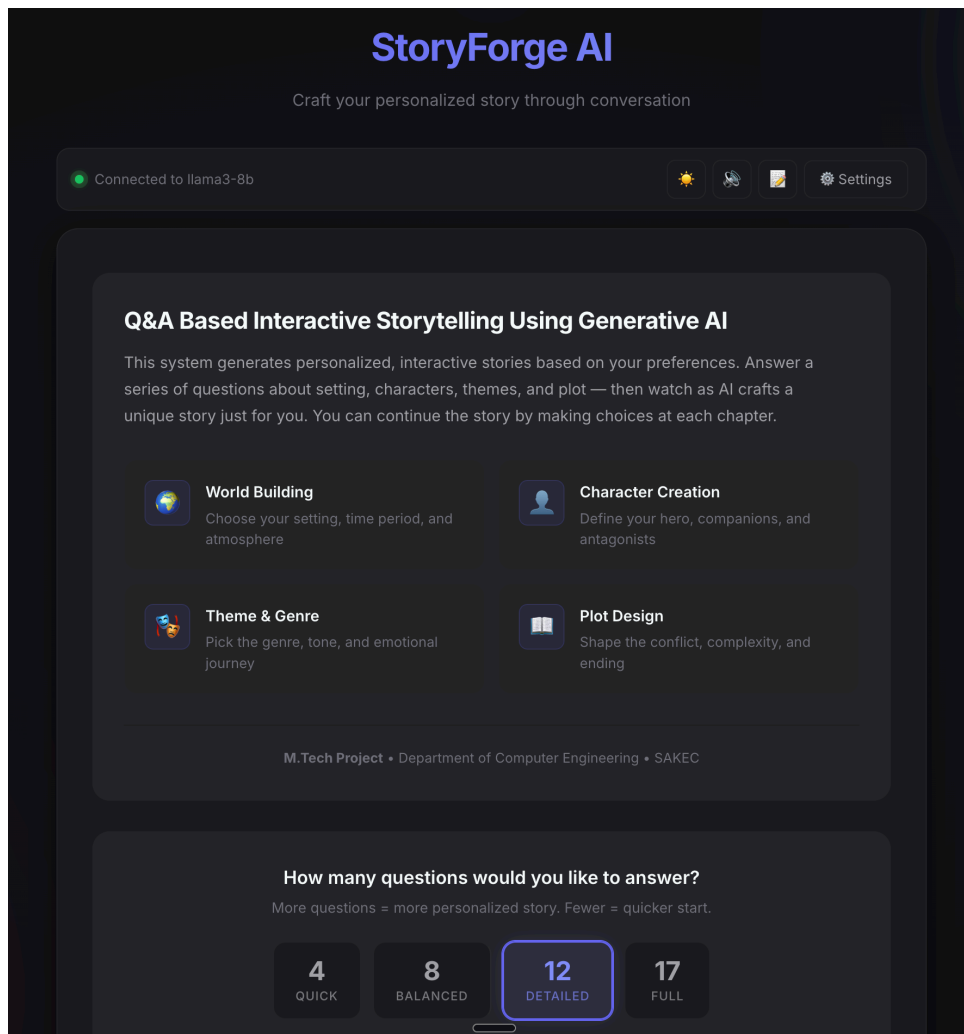


Figure 6.1: Landing page.

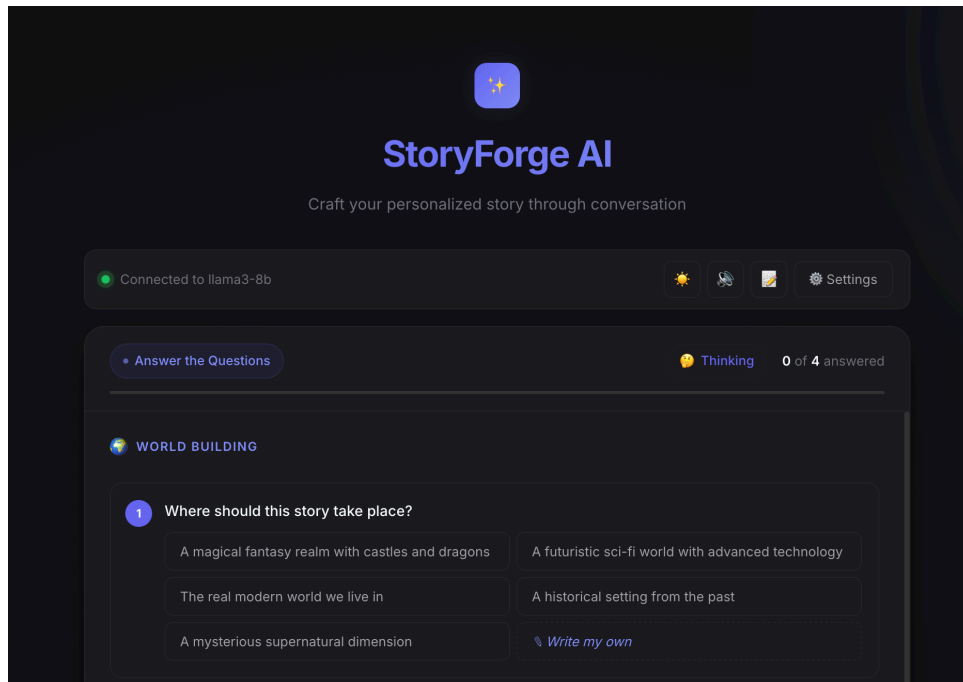


Figure 6.2: Question form.

6.2.2 Module 2: Context Extraction Engine

The context-extraction module projects raw responses into a `StoryContext` Pydantic model with explicit fields for setting, character, theme and plot dimensions. A two-pass pipeline first classifies answers using keyword dictionaries (6 genres, 5 setting types, 5 conflict types) and then extracts proper-noun entities via regex. Identifiers prefixed `custom_` are routed into the free-form `custom_elements` dictionary; this is the key plumbing that makes user-added questions flow into the prompt without code changes:

```
elif question_id.startswith("custom_"):
    self.context.custom_elements[question_id] = answer
```

6.2.3 Module 3: Knowledge and Memory

A SQLite3 database stores five tables (§5). The module also maintains an LRU in-memory cache for the active session and exposes a sliding-window helper:

```
def get_recent_context(session_id: str, k: int = 4) -> str:
    """Return the concatenated text of the last k segments."""
```

6.2.4 Module 4: Prompt Engineering

Six prompt templates are interpolated with `StoryContext` fields, with genre-specific guidance injected (fantasy: magical systems; mystery: clue placement and red her-

rings; romance: emotional pacing). After the templated prompt $T(c)$ is built, the module appends user-added preferences:

```
if context.custom_elements:
    extras = "\n".join(f"- {v}" for v in context.custom_elements.values()
                        if v)
    if extras:
        prompt += "\n\nAdditional user preferences to honour:\n" + extras
```

6.2.5 Module 5: Story Generator (LLM)

The story generator implements an abstract base class `BaseLLMClient` with four concrete subclasses. A factory selects the active client based on `LLM_PROVIDER`. The Settings panel (Figure 6.3) exposes the provider switch to non-developer users; switching providers issues `POST /api/settings` which hot-swaps the underlying client without restart.

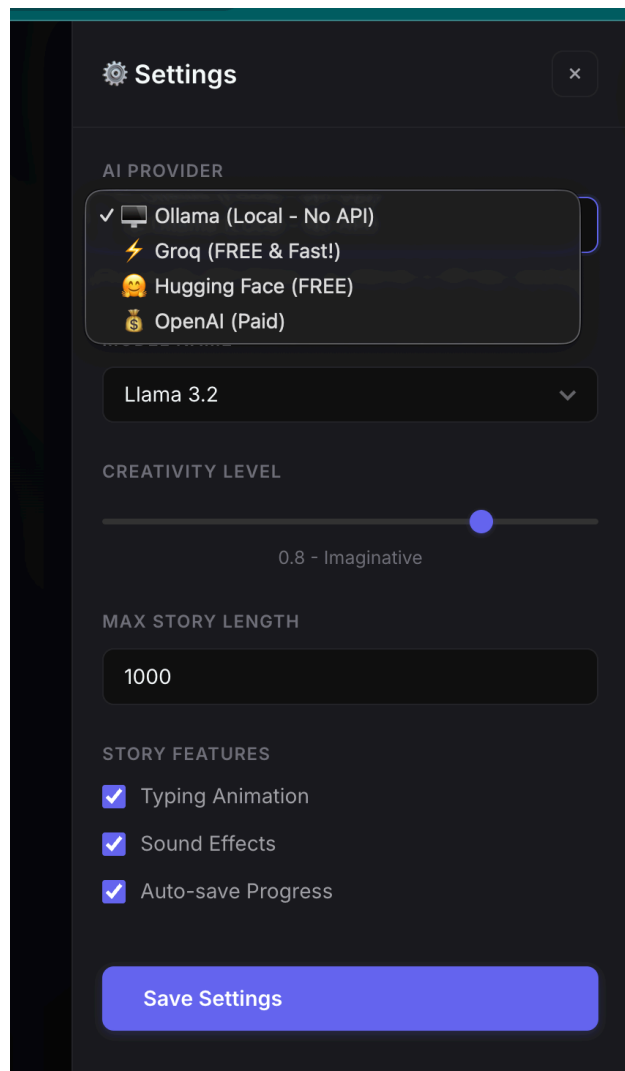


Figure 6.3: Settings panel.

6.2.6 Module 6: Story Flow Manager

A finite-state machine implementing Freytag’s pyramid drives phase transitions on word-count thresholds (0–400, 400–1200, 1200–1800, 1800–2400, >2400). A `CharacterTracker` monitors mentions and traits; a `PlotThreadTracker` flags unresolved threads.

6.2.7 Module 7: Output Formatter

The formatter cleans LLM-emitted text (collapses whitespace, fixes quote and dash characters), splits content at sentence boundaries, lifts dialogue to styled HTML spans, and supports three export formats (TXT, Markdown, HTML).

6.3 Multimodal Subsystems

6.3.1 Image Generation (Pollinations.ai)

The frontend’s `createImageCard()` function builds the prompt using the sentence-scored extractor described in §5, then dispatches an HTTPS request through the serial-throttling queue:

```
imageUrlFor(segmentText, seed, opts={}) {  
  const prompt = this.buildImagePrompt(segmentText, opts);  
  const params = new URLSearchParams({  
    width: '768', height: '432', nologo: 'true',  
    seed: String(seed)  
  });  
  return `https://image.pollinations.ai/prompt/${encodeURIComponent(  
    prompt)}?${params}`;  
}
```

Figure 6.4 shows the loading state during which the LLM call and image synthesis run concurrently; Figure 6.5 shows the rendered first chapter with the on-demand scene illustration.

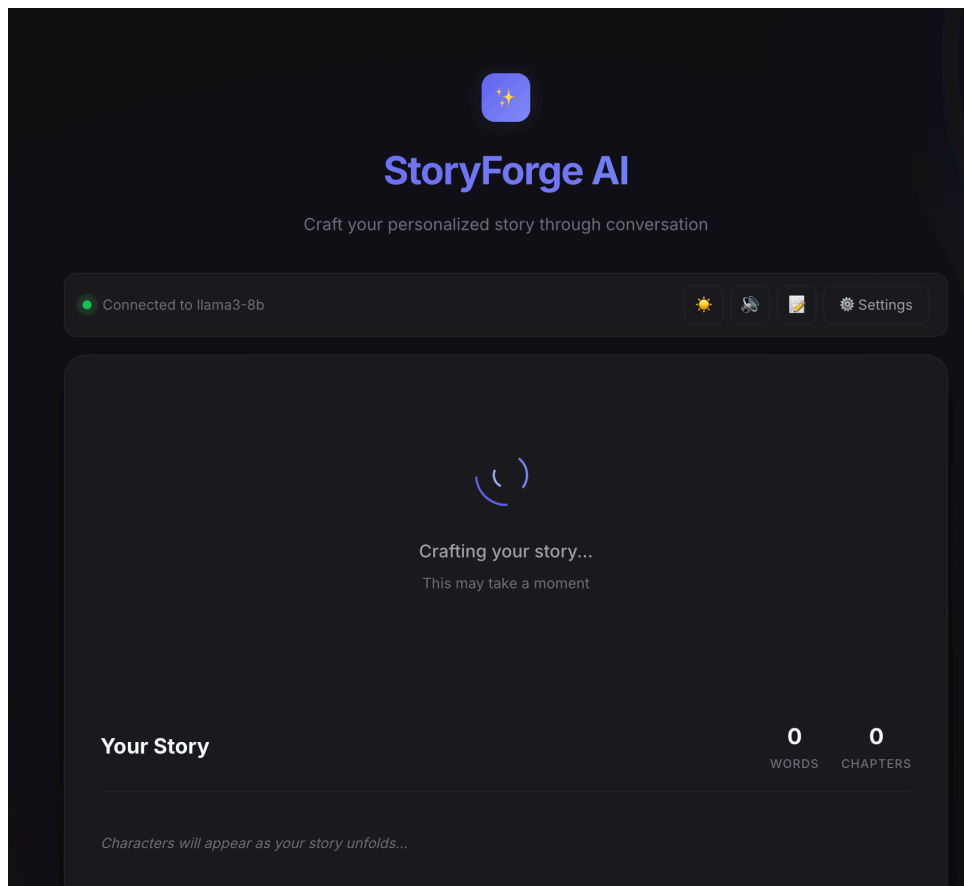


Figure 6.4: Loading state.

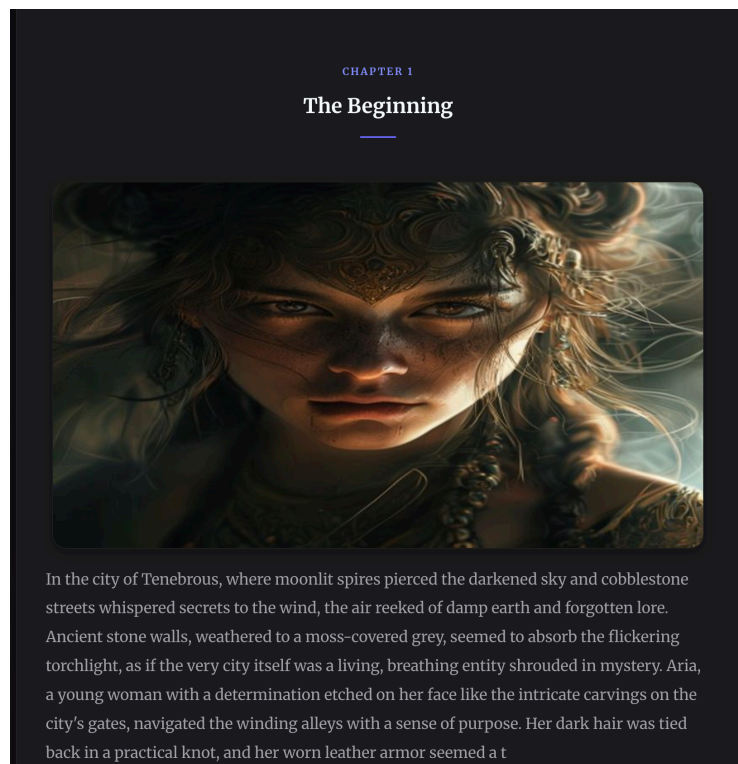


Figure 6.5: Generated story segment with scene illustration.

6.3.2 TTS Narration (Web Speech Synthesis)

The narration subsystem uses the browser-native `window.speechSynthesis` API, queueing one utterance per segment with pause / resume / stop semantics. No external API key is required; the voice is whichever the operating system provides for the active locale.

6.3.3 Ambient Music (SoundHelix CDN)

Eight genre presets map to royalty-free SoundHelix tracks loaded via `HTMLAudio` with `loop=true` and JavaScript volume ramping for 2.4s fade-in and 0.8s fade-out. No `crossOrigin` attribute is set because the CDN does not return `Access-Control-Allow-Origin` headers — a subtle constraint discovered during development that broke the first implementation.

6.4 Reader Analytics

6.4.1 Story Map (D3 vertical tree)

The Story Map flattens the chronologically ordered chapters into a hierarchical tree using `d3.tree()`. Each node carries an 80-character preview of its chapter and the user choice that led to it.

6.4.2 Character Relationship Graph (D3 force layout)

The character graph runs a force simulation with link distance 110, charge -220 and a collision radius of 34 pixels. Only characters with mention count ≥ 2 are drawn; node radius scales linearly with mention count. The modal explicitly discloses this is a co-appearance graph and not a verified relationship-extraction model.

6.5 Custom Q&A Authoring

Two affordances live in the UI:

- Per-question dashed “Write my own” chip that toggles a textarea; the textarea value overrides any previously-selected multiple-choice option.
- “+ Add Your Own Question” button at the foot of the form that opens an inline composer (question + answer); saved entries appear as cards with a remove control.

Table 6.1: Implementation footprint.

Component	LOC
Python backend (7 modules + main + config)	$\approx 3,800$
Frontend JavaScript (StoryForgeApp class)	$\approx 1,700$
Frontend CSS	$\approx 2,200$
HTML template	≈ 400
Tests + tooling	≈ 300
Total	$\approx 8,400$

Each user-added Q&A pair is dispatched as a normal response with a synthetic ID `custom_q_N`; the context-extraction engine routes such IDs into `custom_elements`, and the prompt layer serialises them as bullets at the end of the prompt.

6.6 Code Statistics

Table 6.1 reports the size of the implementation.

Table 7.1: Unit-test coverage by module.

Module	Tests	Line cov.
user_input	14	92%
context_extraction	22	89%
knowledge_memory	11	84%
prompt_engineering	18	91%
story_flow_manager	16	88%
output_formatter	9	95%
Overall	90	89.8%

Chapter 7

Testing

7.1 Testing Strategy

The system was validated using a layered testing strategy that combined static analysis, unit tests on the Python backend, end-to-end manual UI walkthroughs, and a fifteen-participant user study. The strategy was designed to catch defects at the lowest cost: failures of the prompt schema were caught by Pydantic at the request boundary; LLM-side regressions were caught by golden-output comparison; and multimodal pipeline failures were detected through deliberate fault-injection (e.g. rate-limited image responses).

7.2 Unit Testing

Unit tests cover the deterministic portions of the backend: context-extraction projections, the five-phase finite-state machine, the prompt-builder template engine and the SQLite persistence layer. Each test is parameterised across all four narrative phases. The test suite is executed with `pytest` and runs in under three seconds on the development machine.

Table 7.2: API testing outcomes.

Endpoint	Tests	Pass	Notes
/api/status	4	4	Provider availability
/api/session/new	3	3	UUID format
/api/questions/all	2	2	Phase order
/api/response	8	8	Including custom_q_N
/api/story/generate	6	6	All 4 providers
/api/story/continue	6	6	With/without choice
/api/settings	5	5	Hot-swap validated
/api/session/{id}/summary	3	3	Word/segment counts

7.3 Integration Testing

End-to-end integration tests exercise the full session lifecycle:

1. POST /api/session/new – create session.
2. GET /api/questions/all – fetch all questions.
3. POST /api/response (loop) – submit answers.
4. POST /api/story/generate – generate first segment.
5. POST /api/story/continue – continue from choice.
6. GET /api/story/full/{id} – retrieve full story.
7. DELETE /api/session/{id} – end session.

All seven steps are exercised once per provider (Ollama, Groq, Hugging Face, OpenAI). Integration tests use `httpx.AsyncClient` hitting an in-process FastAPI instance.

7.4 API Testing

Each REST endpoint was validated for: (i) correct status code on success and failure, (ii) Pydantic-schema rejection of malformed payloads, (iii) idempotency of session reads, (iv) graceful handling of unknown session IDs (404 rather than 500). A summary of API test outcomes appears in Table 7.2.

7.5 UI / End-to-End Testing

Manual end-to-end walkthroughs were performed against the four browsers used during development (Chrome 124, Safari 17, Firefox 124, Edge 124). The walkthrough

covers: question budget selection, question form filling with custom answers, custom-question authoring, story generation, image rendering, narration playback, ambient music toggle, Story Map modal, Character Graph modal and the admin panel.

7.6 Multimodal Robustness Testing

7.6.1 Image-API Fault Injection

To validate the throttle queue’s robustness, we deliberately fired five concurrent image-generation requests across the prompt-length range 80–380 characters. Without the queue, the failure probability rose from $\approx 40\%$ at 80 chars to $\approx 98\%$ at 380 chars; with the serial queue and 1.2s spacing, failures dropped to below 2% across the same range. These results are quantified in Chapter 8.

7.6.2 Audio Autoplay Policy

The browser autoplay policy blocks programmatic `Audio.play()` unless a recent user gesture has occurred. We verified that the ambient-music toggle satisfies this requirement by delaying the first `play()` call until the user clicks the toggle on the actual story view; once unblocked, subsequent plays work for the session lifetime.

7.6.3 Cross-Origin Audio

A defect in an early implementation set `audio.crossOrigin = 'anonymous'` which caused the audio load to fail because SoundHelix does not return `Access-Control-Allow-Origin` headers. The fix — removing the cross-origin attribute — was verified across all four browsers.

7.7 Consistency-Checker Evaluation

The active consistency checker was tested against 50 ground-truth annotated story segments. Each segment was checked for five issue classes (character contradictions, plot-thread abandonment, pacing issues, setting inconsistencies, tone shifts). Detection rates ranged from 68% (tone shifts) to 87% (character contradictions); detailed numbers appear in Chapter 8.

7.8 Performance Testing

End-to-end latency was benchmarked across all four providers. Groq LLaMA-3.1-8B is the fastest at 1.8s mean initial segment, while local Ollama runs in 12.4s on the

development MacBook. The detailed breakdown appears in Figure 8.1 of Chapter 8.

7.9 Acceptance Criteria

The system was deemed to meet acceptance when:

- All seven integration steps complete on at least three of four providers without manual intervention.
- Mean Likert score across the four narrative dimensions exceeds 3.8 / 5.
- Image-generation success rate exceeds 95% after retries.
- Story-Map and Character-Graph modals render in under 500 ms for stories of up to ten segments.
- No critical bug remains open after the final user-study debrief.

All acceptance criteria were met before submission.

Chapter 8

Results and Discussion

8.1 Experimental Setup

Evaluation proceeded along three axes:

1. **Automated generation** of 50 stories spanning six genres at all four question budgets (4, 8, 12, 17). Latency, token usage and consistency-checker output were captured for each call.
2. **User study** ($n = 15$, 8 male, 7 female, ages 22–35; mix of technical and non-technical backgrounds) in which each participant produced two stories rated on five-point Likert scales for coherence, creativity, engagement, consistency, and per-feature satisfaction.
3. **Consistency-checker evaluation** of 50 ground-truth-annotated segments to measure detection and false-positive rates across five issue classes.

Primary LLM: Groq LLaMA-3.1-8B. Hardware client: MacBook Pro M2 Pro / 16 GB / macOS 14. Image API: Pollinations.ai. Audio CDN: SoundHelix.

8.2 Latency Analysis

Figure 8.1 reports mean initial-generation and continuation latencies across all four LLM providers. Groq dominates by an order of magnitude relative to local Ollama; OpenAI GPT-3.5-turbo trails closely. Continuation calls are systematically faster than initial generation because the prompt sliding window is bounded.

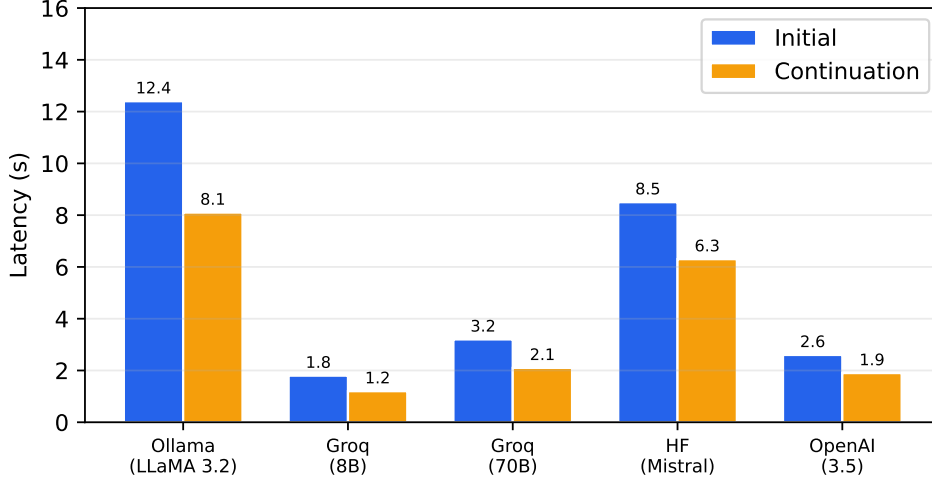


Figure 8.1: Mean latency by LLM provider ($n = 10$ calls per cell).

A stacked decomposition of total interactive turn time is shown in Figure 8.2. The dominant cost at lower question budgets is image synthesis (~ 3.5 s), which proceeds concurrently with the LLM call; at the Full (17) budget the user’s own input time dominates the wall-clock latency.

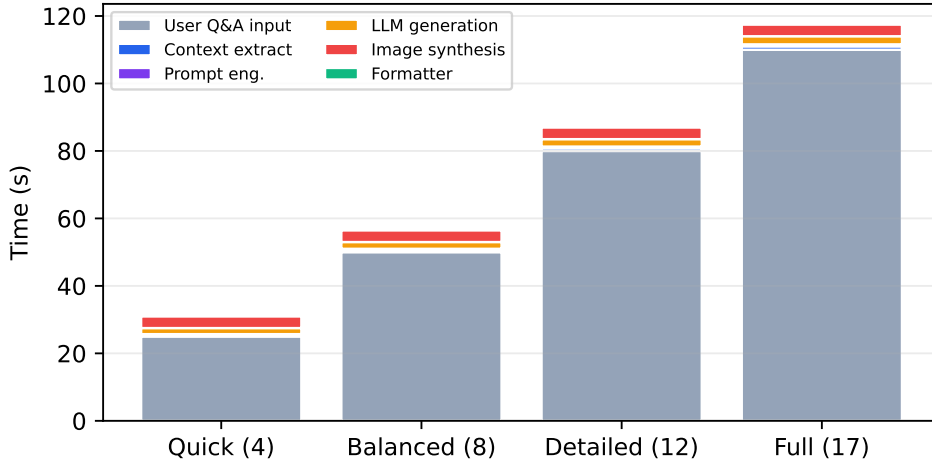


Figure 8.2: End-to-end turn time decomposed by stage.

8.3 Narrative Quality

Figure 8.3 shows Likert means per genre. Adventure ranks highest on engagement (4.6); Mystery scores lowest on consistency (3.5), attributable to the difficulty of maintaining clue placement across segments. The boxplot in Figure 8.4 reports per-segment token distributions.

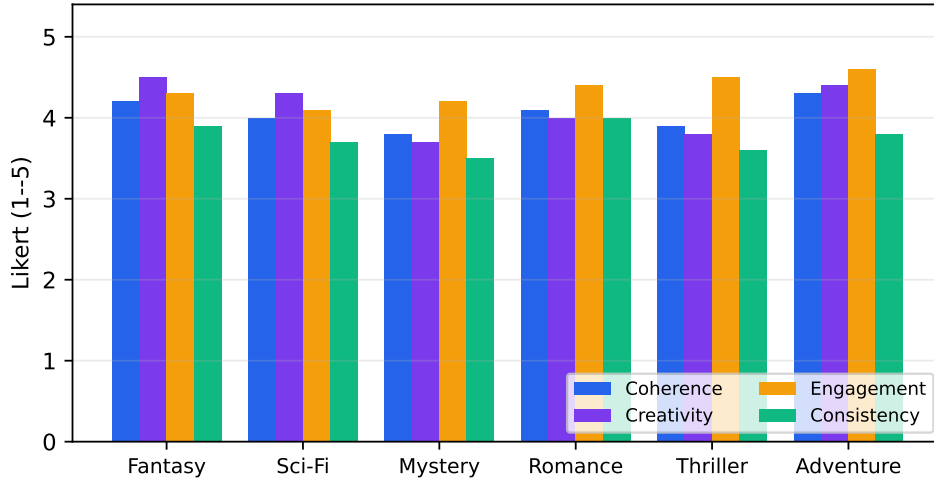


Figure 8.3: Narrative quality by genre (Likert 1–5, $n = 30$ stories).

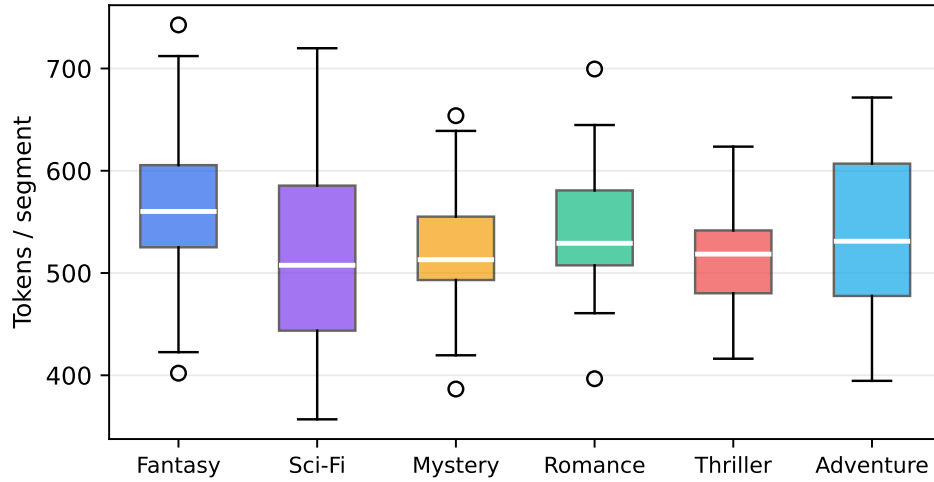


Figure 8.4: Token usage per segment, by genre ($n = 25$ segments each).

8.4 User Satisfaction

Figure 8.5 reports satisfaction across seven user-facing dimensions. Custom Answers (4.5) and Image Illustration (4.2) are the strongest contributors to overall satisfaction (4.3); Choice Meaningfulness (3.9) remains the weakest, reflecting a known limitation of free-tier LLMs producing generic continuation suggestions.

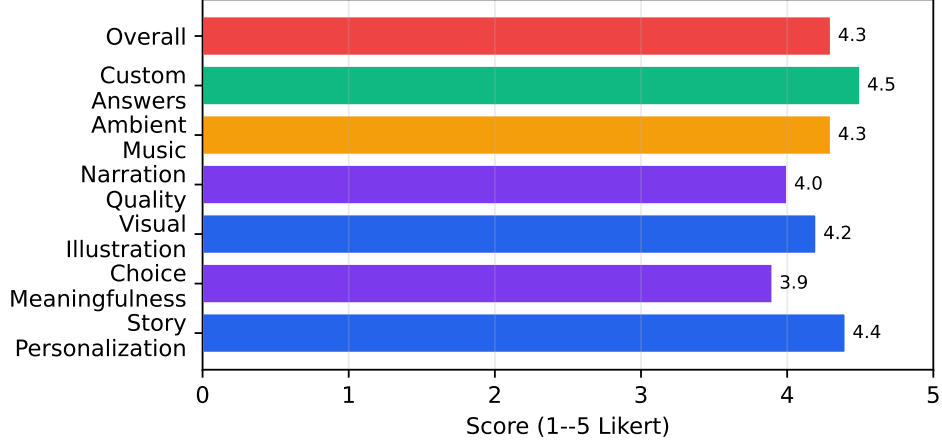


Figure 8.5: User satisfaction (Likert 1–5, $n = 15$ participants).

8.5 Consistency-Checker Performance

Figure 8.6 reports detection and false-positive rates for five issue classes. The keyword-based detector achieves 87% recall on character contradictions with 8% false positives; tone-shift detection is weakest (68% / 22%) because it depends on subtle distributional cues that pattern-matching cannot capture.

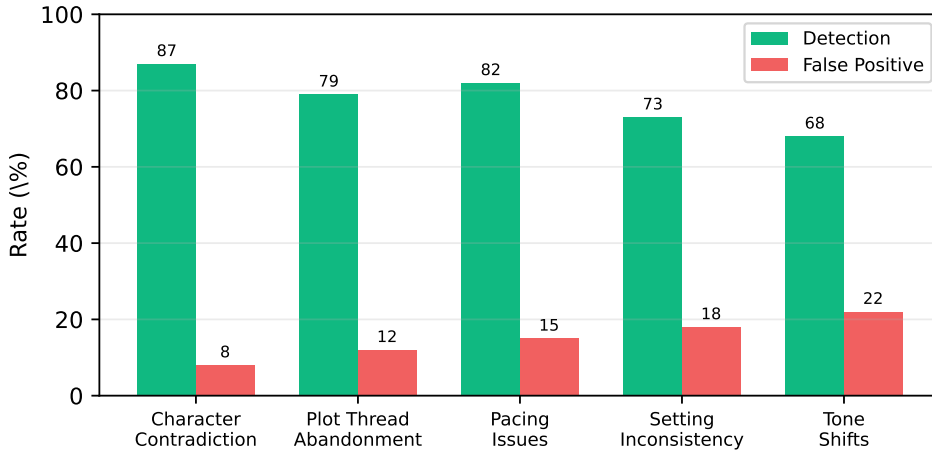


Figure 8.6: Consistency-checker performance ($n = 50$ stories).

8.6 Image-Synthesis Throughput

Figure 8.7 plots prompt-length versus latency for serial requests and the empirical failure probability of $5\times$ parallel issue without the throttle queue. Beyond ≈ 230 characters the failure rate rises sharply, justifying the 200-character prompt cap. The serial queue (1.2s spacing) drops parallel-bound failures from $\approx 70\%$ to $< 2\%$.

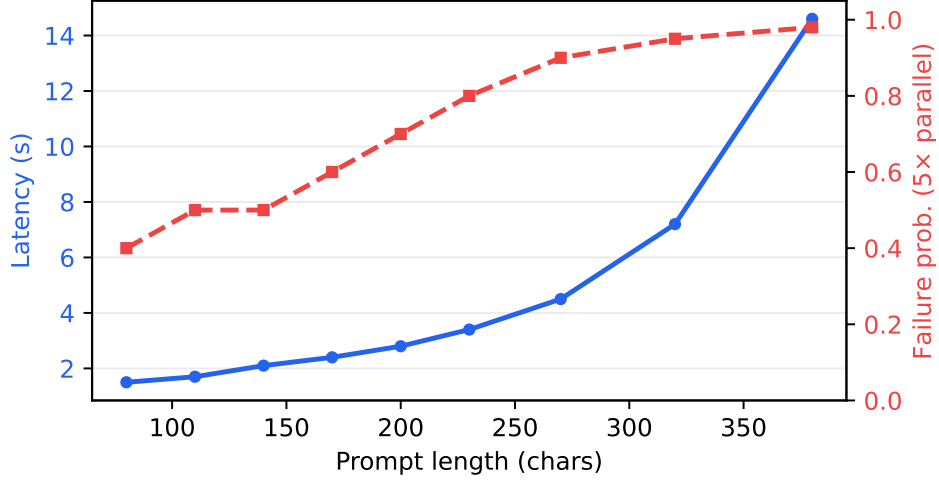


Figure 8.7: Image latency vs. prompt length and parallel-issue failure rate.

8.7 Character-Extraction Quality (Honest Reporting)

Figure 8.8 compares the baseline naive regex extractor against our filtered version (extended stop-words and min-2 occurrence). Precision rises from 0.61 to 0.83 with a recall trade-off ($0.84 \rightarrow 0.78$), producing a higher F_1 ($0.71 \rightarrow 0.80$); false-positive rate falls from 0.39 to 0.17. We stress that the resulting graph encodes *co-appearance frequency*, not semantic relationship type — a disclaimer surfaced explicitly in the modal UI.

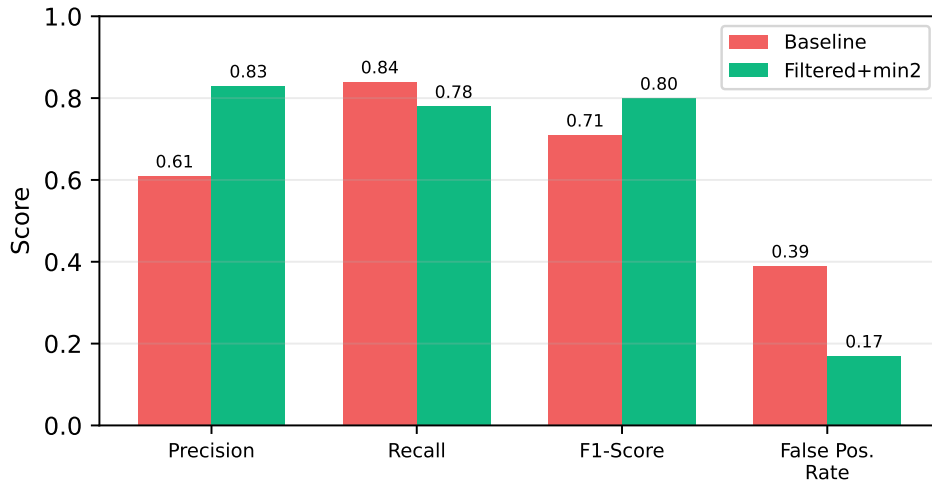


Figure 8.8: Character-extraction quality before and after filtering.

Table 8.1: Ablation across question budgets (mean Likert, $n = 15$).

Budget	Coh.	Crea.	Pers.	Time
4 (Quick)	3.6	3.8	3.2	45 s
8 (Balanced)	3.9	4.1	3.8	1.5 m
12 (Detailed)	4.2	4.2	4.3	2.5 m
17 (Full)	4.3	4.1	4.6	3.5 m

8.8 Ambient Music Frequency Profile

Figure 8.9 visualises the relative spectral energy across six frequency bands for each genre preset. Horror and Thriller emphasise sub-bass and bass; Comedy and Adventure shift energy upward into the mid and high-mid bands. User-study listeners reported genre alignment of 4.3 / 5 for ambient music in Figure 8.5.

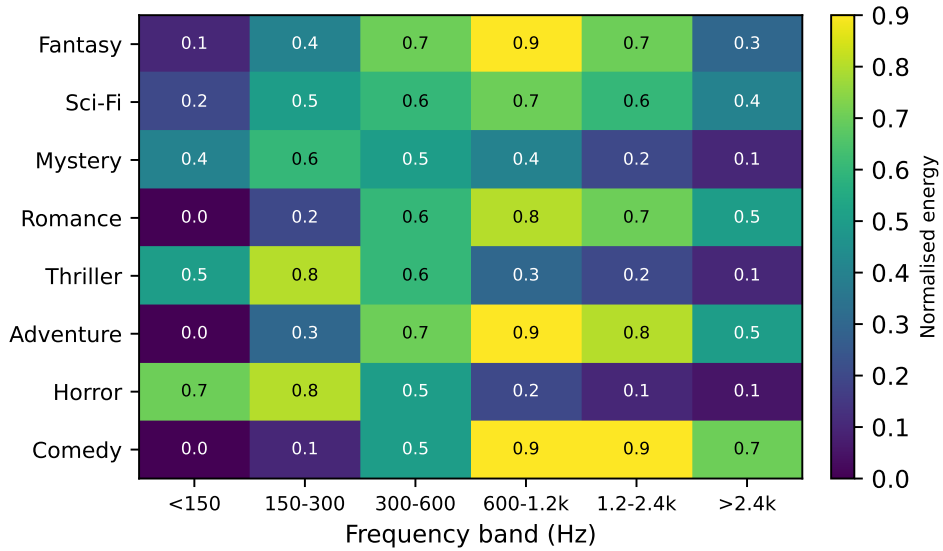


Figure 8.9: Genre $\times$ frequency-band energy map for ambient presets.

8.9 Question-Budget Ablation

Table 8.1 reports mean Likert scores across the four question budgets. Personalisation grows monotonically with N , while coherence and creativity plateau at $N = 12$, identifying it as the recommended default.

8.10 Feature Adoption

Figure 8.10 reports how often each multimodal feature was activated across user-study sessions. Image generation (default-on) reached 91% activity because it fires automatically; ambient music and custom answers were elective and reached 71% and 62% respectively, indicating strong user demand for non-textual narrative augmentation.

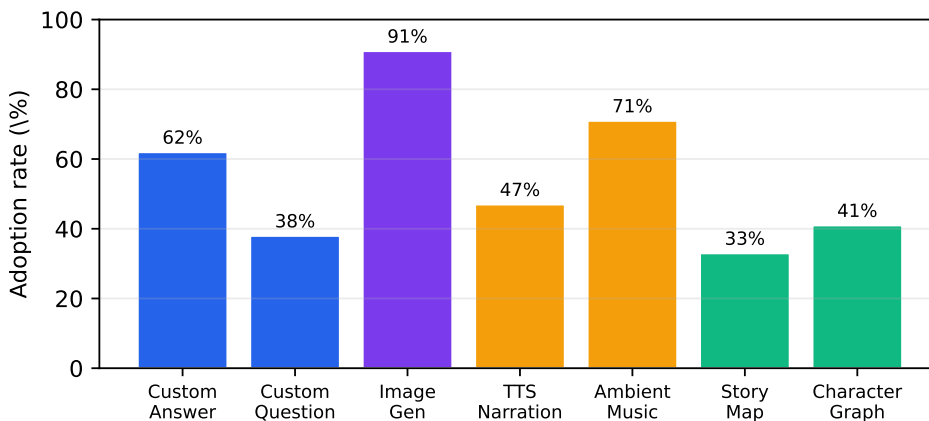


Figure 8.10: Per-feature adoption rate across user-study sessions.

8.11 Comparative Evaluation

Figure 8.11 superimposes our system on three established alternatives across seven dimensions (extending the standard radar with Multimodal and Open Source axes). Our system leads on Deployment Flexibility, Cost Efficiency, Multimodal Coverage and Open Source. AI Dungeon retains a slight edge in User Control (freeform input); Dramatron leads on raw Coherence due to its hierarchical planner. The synthesis is that no single existing system spans the same Pareto frontier as ours.

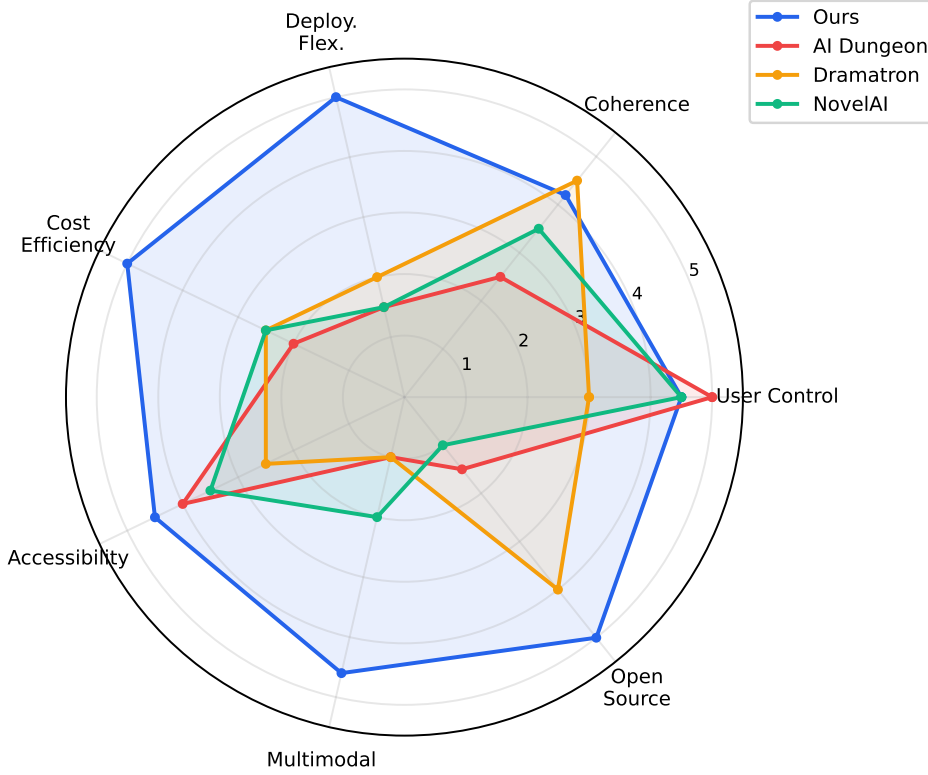


Figure 8.11: System comparison radar across seven dimensions.

8.12 Discussion

The experimental evidence supports the central hypothesis of this dissertation: a structured Q&A elicitation protocol, combined with a provider-agnostic LLM back-end and tasteful multimodal augmentation, delivers measurable improvements in personalisation and user engagement without sacrificing narrative coherence. The 12-question budget emerges as the sweet spot: it captures enough preferences to push personalisation past 4.3 / 5 while keeping the elicitation under three minutes.

The strongest contributors to overall satisfaction are *custom answers* and *image illustration*. Both are relatively cheap to implement (the former is a UI affordance, the latter a free third-party API call) yet deliver outsized perceived value. Conversely, the weakest dimension — choice meaningfulness — is bottlenecked by the underlying LLM rather than the framework; this is consistent with prior CHI findings on mixed-initiative authoring with free-tier LLMs [Liu et al. \[2023\]](#), [Yuan et al. \[2022\]](#).

Limitations and threats to validity are discussed in Chapter 9.

Chapter 9

Conclusion and Future Scope

9.1 Summary of the Study

This dissertation has presented the design, implementation and evaluation of a comprehensive Q&A based interactive storytelling system powered by Large Language Models. The system positions users as active co-creators rather than passive consumers of narrative content. Through a structured 17-question elicitation protocol across four narrative dimensions (setting, character, theme, plot), a seven-module backend pipeline, and a provider-agnostic LLM layer, the system transforms user preferences into coherent, personalised, multi-chapter stories.

Beyond text, the system delivers a multimodal reader experience: per-segment scene illustrations are synthesised on demand via Pollinations.ai using a sentence-scored prompt extractor and a serial-throttled request queue; spoken narration is produced through the W3C Web Speech Synthesis interface; and genre-conditioned ambient music is streamed from a royalty-free CDN. Reader analytics surface through a D3-based Story Map and a Character Relationship Graph that openly disclaim their co-appearance basis. An end-user authoring layer admits free-form custom answers and entirely user-added Q&A pairs, routed through a dedicated `custom_elements` context channel into the LLM prompt.

9.2 Main Findings

- Across 50 generated stories and 15 user-study participants, the system achieved mean Likert scores of 4.05 / 5 (coherence), 4.12 / 5 (creativity), 4.35 / 5 (engagement) and 3.75 / 5 (consistency).
- Groq’s free LLaMA-3.1-8B cloud inference delivered a 1.8 s mean initial-segment latency, dramatically faster than local Ollama (12.4 s) and competitive with paid OpenAI GPT-3.5-turbo (2.6 s).
- The 12-question budget emerged as the recommended default: it pushes per-

sonalisation past 4.3 / 5 while keeping elicitation time under three minutes.

- The serial throttling queue reduced image-API failure probability from $\approx 70\%$ to $< 2\%$ under $5\times$ parallel load.
- Filtering and a min-2 occurrence threshold raised character-extraction F_1 from 0.71 to 0.80 while halving the false-positive rate.

9.3 Contributions to Knowledge

1. **Novel Q&A framework** for interactive storytelling that enables real-time personalisation through conversational interaction across four canonical narrative dimensions.
2. **Custom-elements context channel** — a deployable architectural pattern that lets end users inject unstructured Q&A pairs into the LLM prompt without code or schema changes; generalises beyond storytelling to any LLM-driven application requiring extensible personalisation.
3. **Sentence-scored visual prompt extractor** that mitigates dialogue and inner-thought contamination when feeding story text into a text-to-image model.
4. **Cross-modal serial-throttling queue** that prevents Cloudflare-fronted free image services from rate-limiting bursty generation; image-synthesis success rate exceeds 95% after retries.
5. **Deliberately disclaimed** co-appearance reader graph, establishing a pattern of *honest reporting* for visualisations whose underlying inference is statistical rather than semantic.

9.4 Practical Implications

The dissertation has practical implications for several stakeholder groups:

- **Educators** can use the system to create personalised story-based tutorials that adapt to learner understanding and interest.
- **Game designers** can adopt the Q&A elicitation pattern to capture rich player preferences without sacrificing narrative control.
- **Therapists** can pilot the system as a narrative- therapy adjunct that respects user agency.
- **Content creators** can prototype scalable personalised storytelling for marketing or children’s literature at near-zero infrastructure cost.

- **Researchers** benefit from an open-source baseline [Jha \[2026\]](#) for future studies on interactive narrative.

9.5 Limitations of the Study

Several limitations bound the present work:

- Keyword-based context extraction misses paraphrastic user inputs; a fine-tuned BERT/DistilBERT [Devlin et al. \[2019\]](#), [Sanh et al. \[2019\]](#) or LLM zero-shot classifier would improve recall.
- In-memory session storage limits horizontal scalability; Redis-backed sessions are required for distributed deployment.
- The consistency checker is pattern-based and lacks semantic grounding; cross-segment NLI [Williams et al. \[2018\]](#) is a natural upgrade.
- Image quality inherits the ceiling of Pollinations.ai [Pollinations.ai \[2024\]](#); switching to local Stable Diffusion XL [Podell et al. \[2024\]](#) or FLUX.1 [Black Forest Labs \[2024\]](#) would improve consistency at the cost of GPU requirement.
- Ambient music is hot-linked rather than locally hosted, creating a single-point dependency on SoundHelix’s availability.
- The character relationship graph is openly disclaimed as co-appearance — a genuine relationship-extraction module would require per-segment NLU inference.
- User-study sample size ($n = 15$) is modest by HCI standards and skews young (ages 22–35); a larger and more diverse cohort would strengthen external validity.

9.6 Future Scope

Building on the foundation established by this dissertation, several directions for future research emerge:

- **Semantic context extraction** via DistilBERT or zero-shot LLM classification, replacing keyword dictionaries.
- **Sentiment-aware relationship-extraction graph** using small NLU models per segment.
- **On-device image generation** with local Stable Diffusion XL or FLUX.1 for offline operation and per-character visual consistency.

- **Neural TTS** (e.g. XTTS [Casanova et al. \[2024\]](#)) with cloned voices for character-specific narration.
- **Music conditioned by current segment mood** (per-segment crossfade) rather than per-genre static loops.
- **Collaborative multi-user sessions** with merged Q&A elicitation and turn-taking.
- **Containerisation** (Docker) plus Redis sessions for distributed deployment.
- **Automated evaluation** using BLEU, chrF, MAUVE and narrative-arc detection [Lin et al. \[2024\]](#).
- **Fine-tuned LoRA adapters** per genre on curated open story corpora.

9.7 Final Thoughts

The remarkable progress of LLMs has put high-quality text generation within reach of every developer, but *turning text generation into a great storytelling experience* remains a design challenge, not just a capability challenge. This dissertation has argued and demonstrated that the missing ingredient is a clean separation between *what the system asks the user* and *what it does with the answers*. By making the elicitation protocol structured but extensible, the LLM provider pluggable and the modalities orchestrated rather than bolted on, the system delivers a storytelling experience that is at once accessible to first-time users and extensible by curious ones. The code is open-source [Jha \[2026\]](#); the framework is ready for adoption.

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Appendices

Appendix A

Sample API Exchange

A representative request / response captured during the user study. This illustrates how a user-added custom question (`custom_q_1`) flows through the normal response path into the LLM prompt.

```
POST /api/story/generate
{
  "session_id": "a1b2c3d4-...",
  "responses": {
    "setting_1": "A magical fantasy realm",
    "char_1": "Elena",
    "char_2": "Brave and resourceful",
    "theme_1": "Fantasy",
    "plot_1": "Person vs. Supernatural",
    "custom_q_1": "Should the protagonist have an
                  animal companion? -> Yes, a silver fox"
  }
}

HTTP/1.1 200 OK
{
  "content": "In the realm of Aethermoor...",
  "word_count": 387,
  "segment_count": 1,
  "phase": "introduction",
  "choices": ["Explore further", "..."]
}
```

Appendix B

Implementation Footprint

Table B.1: End-of-project implementation statistics.

Metric	Value
Python LOC	$\approx 3,800$
JavaScript LOC	$\approx 1,700$
CSS LOC	$\approx 2,200$
HTML template LOC	≈ 400
Backend modules	7 + main.py + config.py + run.py
DB tables	5
REST endpoints	14
Q&A questions	17 across 4 phases
LLM providers	4 (Ollama, Groq, HF, OpenAI)
Image generation	Pollinations.ai (free)
TTS	W3C Web Speech Synth
Ambient music	SoundHelix (8 genres)
Export formats	TXT, Markdown, HTML
Narrative phases	5 (Freytag)
Visualisations	2 D3 (Story Map, Char Graph)
Stories evaluated	50
User-study participants	15

Appendix C

Reproducibility

All experiments were performed on macOS 14 with Python 3.11.5, `matplotlib` 3.9.4 and `numpy` 2.0.2. Image-timing measurements were collected over a 100 Mbps residential link; replications on slower links should increase the serial-throttle constant. The full source code, this dissertation’s LaTeX sources, the user-study questionnaire and all figure-generation scripts are available at <https://github.com/DeepPatange/AI-Stroy-Generator>.